Applied Deep Learning – Assignment 1: Initiate

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1 Introduction

Flow Cytometry (FCM) is used in medicine to find surviving cancer cells (blasts) in leukemia patients after treatment [1, 2]. Counting this tiny fraction of blasts is called Measurable Residual Disease (MRD) assessment [1, 2]. Doing this manually is not only time-consuming, but it is also expensive to train experts, and in some regions of the world trained personnel are simply not available. Additionally, deciding whether cells are blasts or not during manual gating is partly subjective, so deep learning is used to automate the process and make it more objective [1, 2].

In 2022, Wödlinger et al. [1] built Flowformer for B-ALL data, which works well overall (median $F_1 \approx 0.94$) but loses accuracy on low-MRD samples (below 0.1% / fraction 0.001 down to 0.01%) [1]. In 2024, Weijler et al. [2] extended this line of work with the FATE architecture. In a recent BSc thesis at TU Wien [3], Normalizing Flows [4, 6] on B-ALL data trained only on healthy cells were tested. No supervised prediction head was trained and the outcome of the model used in the thesis was a higher rate of false positives than the SOTA [3].

2 Topic and Project Type

Topic: Cell classification and MRD prediction in Flow Cytometry data (similar to point-cloud segmentation) [1, 2].

Project Type: Bring your own method. I am taking the existing FATE [2] and Normalizing Flow [3] codebase, adding a supervised training stage for the prediction head on the B-ALL dataset, and testing if the results and F_1 score can be improved.

3 Project Summary

3.1 4a: Project Idea and Approach

In the thesis [3], a Normalizing Flow [4] (8 blocks of Coupled Rational Quadratic Splines [5, 6]) was used to learn the distribution of healthy cells on B-ALL data. To find cancer cells, a threshold on the Negative Log-Likelihood (NLL) anomaly score was used [3]. Without supervised training of a classification head, the model produces too many false alarms on low-MRD samples [3].

The goal is to train a supervised prediction head on top of the pretrained Normalizing Flow latent space [3] (`emb_dim = 128`) on the B-ALL data. My hypothesis is that this pushes the predicted MRD values much closer to the diagonal line ($y = x$) of the predicted-vs.-true MRD plot used for evaluation in [1, 3], especially in the low-MRD regime (below 0.001 / 0.1% down to 0.01%).

To test whether the unsupervised Flow already captures enough information in its latent space or needs further adaptation, I will implement and compare two ideas:

- **Setup 1 (Raw markers, frozen Flow backbone):** Take the Normalizing Flow pretrained on healthy B-ALL cells, freeze the Flow weights, and only train the binary classification head on the latent vectors z [4, 3] using Cross-Entropy loss. This directly tests whether the information needed to separate blasts is already present in the learned latent space.
- **Setup 2 (Raw markers, end-to-end fine-tuning):** Take the pretrained Flow, attach the prediction head, and unfreeze the whole network during supervised training [3]. For the loss function in this mode, I will explore what fits best so that healthy cells stay inside the standard normal distribution while blasts are actively pushed away.

After training, the main evaluation metric will be the F_1 score (alongside recall, precision, and the sample-level MRD plots), which I will compare against the threshold baseline from the BSc thesis [3] and the supervised reference models [1, 2].

3.2 4b: Dataset Description

The data comes from the Computer Vision Lab (TU Wien) and the St. Anna Children's Cancer Research Institute [1, 3]. I will get access to the server and use the same B-ALL dataset as in the BSc thesis [3]. Each sample is stored as a preloaded `.pkl` file with the cell measurements and a second `.pkl` file with the expert blast labels. Only intact cells (`Intact` gate) are used [3].

- **B-ALL Dataset:** 659 samples (397 train, 130 validation, 132 test) [3] with 9 dense markers measured in every sample (`FSC-A`, `SSC-A`, `CD38`, `CD34`, `CD45`, `CD10`, `CD19`, `CD20`, `SY41`) [1, 3]. Using the exact same feature set and data split allows a direct comparison of the supervised head against the unsupervised thesis results [3]. Anonymized test samples may also be used for the final demo application.

3.3 4c: Work-Breakdown Structure

Task	**What I will do**	**Time**
Dataset collection	Get server access and the `.pkl` files, check splits and marker lists.	1 h
Designing & building network	Set up the repo, connect the prediction head and loss functions to the NF latent space.	10 h
Training & fine-tuning	Pretrain/load NF, train the head (Option 1 & 2), test losses, tune hyperparameters.	25 h
Building demo application	Build a Streamlit web app (User uploads an anonymized sample and sees the MRD prediction and scatter plots).	10 h
Writing the report	Write the Report.	8 h
Preparing the presentation	Make the slides and preparing for the presentation.	5 h
Total Project Time	*(Plus 16 h lecture = 75 h total workload)*	**59 h**

References

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